



Buying Tips

✓ **DIGITAL/OPTICAL ZOOM:** Go for a camera with optical zoom if you're going to use it more outdoors than indoors. Don't get carried away by digital zoom.

✓ **MEMORY:** Most cameras have 16 MB of memory or less. This can store some 30-35 images depending on the subject, but if you're going to be traveling for long periods of time, you'll need much more than 16 MB.

✓ **MANUAL SETTINGS:** Buy a camera that has manual exposure and shutter speed settings. This will allow you to take decent images indoors, where light is usually poor.

✓ **BUNDLED CABLES:** Check for necessary cables such as the USB and video cables. Also, go for a camera that doesn't have a proprietary USB cable—these are costly and difficult to procure.

✓ **AESTHETICS:** Buy a camera that offers good grip and has proper placement of buttons. Slim cameras look good, but they are poor as far as ergonomics and functionality goes.

✓ **RECHARGEABLE BATTERIES:** Digital cameras, especially those with built-in LCD screens and optical zoom, need a lot of power. Buy cameras that have rechargeable batteries.

✓ **ENVIRONMENT PROTECTION:** Some cameras offer splash-resistant packaging, which ensures that the camera is not damaged by water or moisture.

- 1 Light passes through the lens and is directed to the CCD.
- 2 The image is focused on the CCD.
- 3 The ADC converts the image from the CCD into digital signals.
- 4 The ADC sends data to the DSP that creates the image and compresses it.
- 5 The final image is stored either in the inbuilt flash memory or on a memory stick.



Quick Find

YOU NEED To be able to take photos to compile personal albums or publish to the web

Basic Use

LOOK FOR Between 640 x 480 and 1,024 x 768 resolution and 2 MB memory; perhaps one that doubles as a Web cam

YOU NEED High quality images with plenty of configurability

Inter Use

LOOK FOR 1,280 x 1,024 resolution with at least 2x optical zoom and 8 MB of storage

YOU NEED Very high quality photographs with all the knobs and switches, plus expandability

Prof Use

LOOK FOR 2,048 x 1,536 or higher resolution with special lens options, and a minimum of 16 MB of memory

Do Remember

- CMOS-based sensors for digital cameras are less expensive to manufacture than CCD-based sensors. But the former can't capture motion correctly. Many industry analysts believe that eventually, almost all entry-level digital cameras will be CMOS-based, and that only mid-range and high-end units will use CCDs.
- Today you can buy photo printers that can print directly from a digital camera.

Facts

- CCD technology was initially developed in the 1960s as a memory device to store data, and it was only in 1974 that Fairchild Electronics developed the first imaging device using CCDs, with a format of 100 x 100 pixels.

The Future

The wide acceptance of digital cameras, and the ready availability of software to modify captured images, have prompted newer operating systems such as Windows XP to bundle built-in driver support, making the devices virtually plug-and-play. There are a few features found only in high-end cameras, which will surely be implemented in entry-level cameras soon. This includes the TruePic technology from Olympus, which uses all the pixels even at lower resolutions. The software thus enlarges the image even when it is in raw format on the camera, so as to get a higher interpolated resolution than what's available with current cameras. This will reduce the artifacting that occurs when enlarging photos using image-editing software. Future cameras will also have the lenses specifically designed to suit digital cameras. This is necessary because as the CCD count goes up, the density increases manifold. Even cameras equipped with the best CCDs might not produce a great image if the lens does not let the CCD look through it precisely when zooming in or out.

Today, USB and FireWire are the interfaces of choice for most digital cameras. But as devices from laptops to PDAs, and even cell phones, embrace wireless connectivity to talk with the PC and other mobile devices, the digital camera will surely follow suit.